

Baldivis Secondary College 2016

Year 9 Semester 2 Exam Question/Answer Booklet

Student Name: _____

SOLE UCTIONS

Teacher: _____

Time allowed for this paper

Reading time before commencing work: ten minutes
Working time for paper: one hours

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Mathematics Formula Sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: one non-programmable calculator approved for use in the WACE examinations

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Year 9 Semester 2 Mathematics Exam

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	19	19	20	36	40
Section Two: Extended	3	3	40	52	60
					100

Total _____/88 _____%

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2016*. Sitting this examination implies that you agree to abide by these rules.

2. Answer the questions according to the following instructions.

Write your answers in this Question/Answer Booklet. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that answer, then circle your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

3. When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to the appropriate number of significant figures (normally 2dp) and include appropriate units where applicable.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. The Mathematics Formula Sheet is to be handed in with your Question/Answer Booklet.

Multiple Choice Section:

Identify the choice that best completes the statement or answers the question.

19 multiple-choice questions

2 marks each: 36 marks

Circle the correct answer

Total : _____/36

- 1 A normal die has six faces numbered from 1 to 6.

What is the sample space?

- A {1, 6} B {6}
C {1, 2, 3, 4} **D** {1, 2, 3, 4, 5, 6}

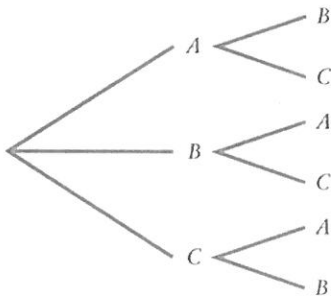
- 2 What is the probability of an impossible event?

- A** 0 B 0.5 C 0.75 D 1

- 3 What is the probability of a certain event?

- A 0 B 0.5 C 0.75 **D** 1

Questions 4 and 5 refer to the following tree diagram. You may assume all branches are equally likely.



- 4 What is the probability of getting B followed by C?

- A $\frac{1}{2}$ B $\frac{1}{3}$ **C** $\frac{1}{6}$ D $\frac{5}{6}$

- 5 What is the probability of getting an A and a B in any order?

- A $\frac{1}{2}$ **B** $\frac{1}{3}$ C $\frac{1}{6}$ D $\frac{5}{6}$

- 6 What is the probability that if a letter is chosen at random from the letters in the word PROBABILITY, it will be a B or an I?

- A** $\frac{4}{11}$ B $\frac{1}{9}$
C $\frac{2}{9}$ D $\frac{2}{11}$

- 7 The probability that Joan will win her race is 0.35. What is the probability she will not win her race?

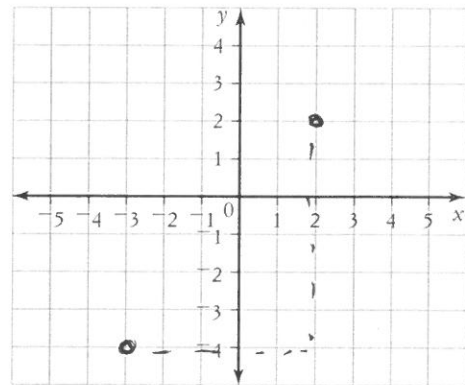
- A 0.35 B 0.5
C 0.65 D 0.75

- 8 What can be said about the gradient of the following line?



- A It is positive. **B** It is negative.
C It is zero. D It is undefined.

- 9 Use the grid below, or the formula, to find the midpoint of the line joining $(-3, -4)$ and $(2, 2)$.



- A $(-1, -1)$ B $(1, 1)$
C $(-\frac{1}{2}, -1)$ D $(\frac{-1}{2}, -1)$

- 10 What is the gradient of the line $x = 3$?

- A 0 B -3
C 3 **D** undefined

- 11 What is the gradient of the line $y = 5$?

- A 5 C -5
B undefined **D** 0

12 What can be said about the gradient of the line below?



- A It is negative. ☐ C It is undefined. ☐
B It is zero. ☐ ☒ D It is positive.

13 Expand $3(2t + 5)$.

- A $5t + 8$ ☐ B $6t + 8$ ☐
C $6t + 5$ ☐ ☒ D $6t + 15$

14 Expand and simplify $(x + 3)(x + 5)$.

- A $x^2 + 3x + 5$ ☐ ☒ B $x^2 + 8x + 15$
C $x^2 + 3x + 15$ ☐ D $x^2 + 15x + 8$ ☐

15 Find the highest common factor of $24ax^3y^2$ and $18axy^2$.

- A $3ax^2y$ ☐ ☒ B $6axy^2$
C $6ax^2y^2$ ☐ D $6ax^2y$ ☐

16 Factorise $4i - 10$.

- A $4(i - 10)$ ☐ B $4(i - 5)$ ☐
C $2(i - 5)$ ☐ ☒ D $2(2i - 5)$

17 Expand $3a(2a - 5b)$.

- A $6a - 5b$ ☐ B $6a^2 - 5b$ ☐
☒ C $6a^2 - 15ab$ ☒ D $6a - 15ab$ ☐

18 Expand $-n(n + 2)$.

- A $-n^2 - 2$ ☐ ☒ B $-n^2 - 2n$
C $-n^2 + 2$ ☐ D $-n^2 + 2n$ ☐

19 Expand $(2t - 4)(5t - 3)$. ☒

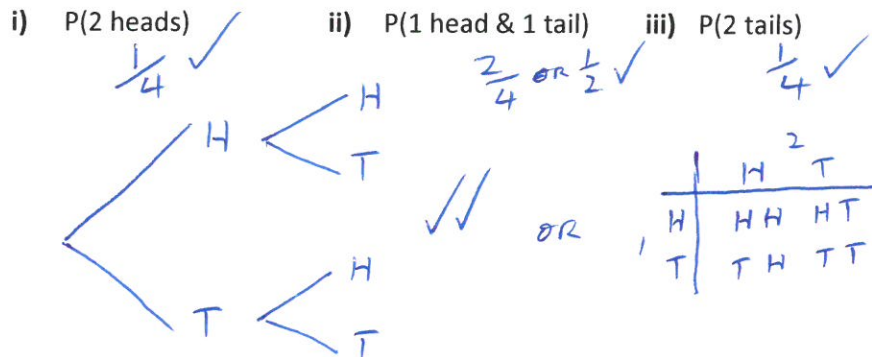
- ☒ A $10t^2 - 26t + 12$ ☐ B $10t^2 - 26t - 12$
C $10t^2 - 14t + 12$ ☐ D $10t^2 + 14t - 12$ ☐

Written Questions (52 Marks)

1.

- a) Assuming that the chances of a head or tail is equal, use a tree diagram or a two-way table to determine the following theoretical probabilities for the tossing of two coins:

(5 Marks)



- b) Determine the following theoretical probabilities of the **totals** when tossing two dice.

(6 Marks)

- i) P(5 or odd) ii) P(5 and odd)
- iii) P(<10 or even) iv) P(odd or divisible by 3)

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

(i) $\frac{18}{36}$ OR $\frac{1}{2}$ ✓

(ii) $\frac{4}{36}$ OR $\frac{1}{9}$ ✓

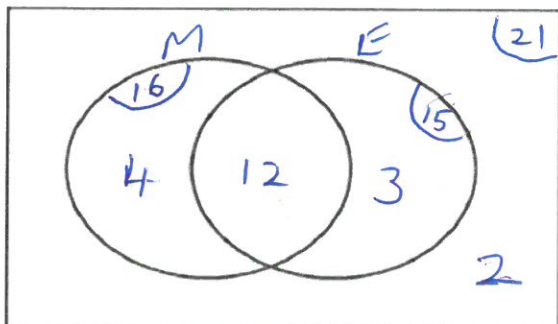
(iii) $\frac{34}{36}$ OR $\frac{17}{18}$ ✓

(iv) $\frac{24}{36}$ OR $\frac{2}{3}$ ✓

- c) In a class of 21 students, 16 students passed Maths, and 15 students passed English. Draw a Venn Diagram and find the probability that a student:

(5 Marks)

- i) passed Maths and English. ii) passed Maths or English. (iii) ~~Failed English~~
- TWO STUDENTS PASSED NEITHER.



i) $\frac{12}{21}$ ✓

ii) $\frac{19}{21}$ ✓

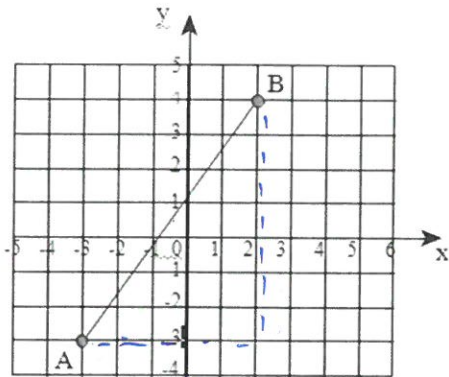
(iii) $\frac{6}{21}$ ✓

2.

a) Find the length, gradient, and midpoint of AB

(4 Marks)

i)



$$AB = \sqrt{5^2 + 7^2}$$

$$= \sqrt{25 + 49}$$

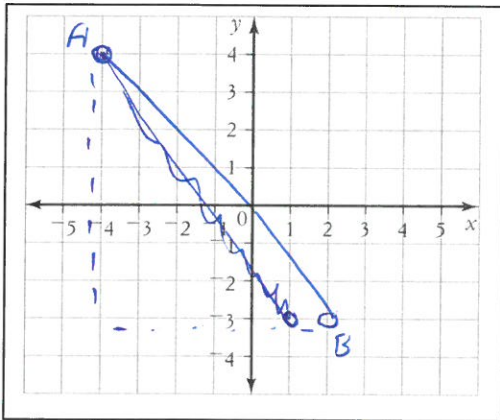
$$= 8.60$$

$$m = \frac{7}{5}$$

$$M_{AB} = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

ii) A(-4,4), B(1,-3)

(5 Marks)



$$AB = \sqrt{6^2 + 7^2}$$

$$= \sqrt{36 + 49}$$

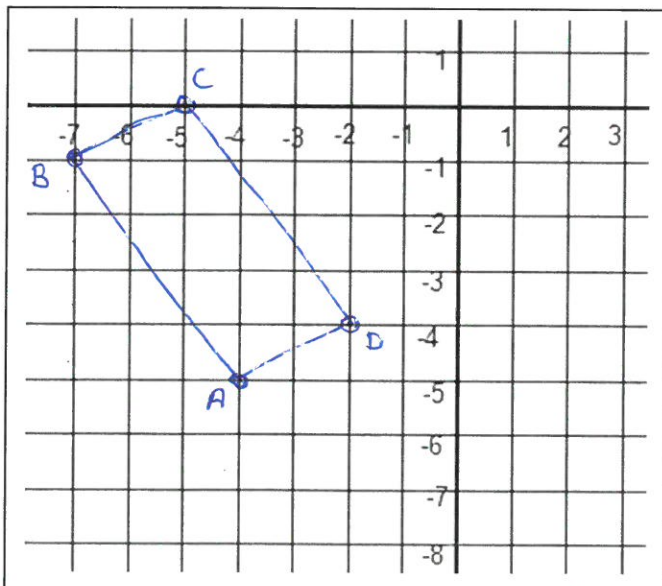
$$= 9.22$$

$$m = -\frac{7}{6}$$

$$M_{AB} = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

b) A parallelogram has the four endpoints A(-4,-5), B(-7,-1), C(-5,0), D(-2,-4).

(7 Marks)



i) Show that AB is parallel to CD (ie have the same gradient).

$$m_{AB} = -\frac{4}{3}$$

$$m_{CD} = -\frac{4}{3}$$

SAME GRADIENT
∴ PARALLEL

ii) Show that the opposite sides are equal in length.

$$AB = \sqrt{1^2 + 3^2} = 5$$

$$CD = \sqrt{4^2 + 3^2} = 5$$

SAME ✓✓

$$BC = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$DA = \sqrt{1^2 + 2^2} = \sqrt{5}$$

SAME ✓✓

3.

a) Simplify the following expressions:

(4 Marks)

i) $3x - 7x$

ii) $5a + 4b - 2a + 2b$

iii) $8b^4 \times (-4b^2)$

iv) $-6d \div 3$

$$= -4x \checkmark$$

$$= 3a + 6b \checkmark$$

$$= -48b^6 \checkmark$$

$$= -2d \checkmark$$

b) Expand each of the following:

(2 Marks)

i) $3(x + 2)$

ii) $-5x(x - 3)$

$$= 3x + 6 \checkmark$$

$$= -5x^2 + 15x \checkmark$$

c) Simplify each of the following by expanding and then collecting like terms:

(8 Marks)

i) $-3(x - 3) + 2(x + 2)$

ii) $(x + 2)(x + 1)$

iii) $(x + 2)(x - 2)$

iv) $(x + 3)^2$

$$= -3x + 9 + 2x + 4 \checkmark$$

$$= -x + 13 \checkmark$$

$$= x^2 + x + 2x + 2 \checkmark$$

$$= x^2 + 3x + 2 \checkmark$$

$$= x^2 - 2x + 2x - 4 \checkmark$$

$$= x^2 - 4 \checkmark$$

$$= (x + 3)(x + 3) \checkmark$$

$$= x^2 + 3x + 3x + 9 \checkmark$$

$$= x^2 + 6x + 9 \checkmark$$

d) Factorize each of the following:

(8 Marks)

i) $4x + 10$

ii) $4t^2 - 12t$

iii) $-2x^3 - 12x$

iv) $x(x - 1) + 5(x - 1)$

$$= 2(2x + 5) \checkmark \checkmark$$

$$= 4t(t - 3) \checkmark \checkmark$$

$$= -2x(x^2 + 6) \checkmark \checkmark$$

$$= (x - 1)(x + 5) \checkmark \checkmark$$